Massive Superstring Spectrum and $D = 11$ SO(10) Structure: Exact Algebraic Derivation and Comprehensive Verification

Massive Modes Analysis Framework

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Abstract

This report presents an exhaustive, ab initio algebraic verification of the Bars conjecture (hep-th/9503228) regarding the emergence of 11-dimensional SO(10) massive supermultiplets in 10-dimensional Type IIA superstring theory. We derive the exact light-cone Green-Schwarz open superstring Fock space up to mass level $n = 10$, factorize the universal 256-state massive supersymmetry factor $\mathcal{M}_{\text{susy}} = \mathbf{128}_B \oplus \mathbf{128}_F$, invert the SO(8) and SO(9) branching rules, and demonstrate that the cumulative spectrum $R_n = \sum_{l=1}^{n} r_l$ lifts into strictly positive integer combinations of SO(10) irreducible representations with zero discrepancy across all 10 levels. All algorithms, mathematical uniqueness proofs, code architecture, and complete term-by-term matching tables are fully documented.

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1 Physics Overview

In 10-dimensional Type IIA superstring theory, the light-cone little group for massless states is SO(8), while the little group for massive particles in $D = 10$ Minkowski spacetime is SO(9). If 10D Type IIA superstring theory is non-perturbatively dual to 11-dimensional M-theory compactified on a circle S^1 of radius R_{11}, the massive states carrying non-zero longitudinal momentum $P_{11} = n/R_{11}$ along the 11th dimension must organize into complete representations of the $D = 11$ massive little group SO(10). In [1], Itzhak Bars proposed that the massive open and closed superstring spectrum, when combined with the non-perturbative Kaluza-Klein and D-brane states carrying momentum along the 11th dimension, assembles into complete $D = 11$ SO(10) supermultiplets.

1.1 Light-Cone Green-Schwarz Oscillators

In the light-cone gauge of the 10-dimensional Green-Schwarz (GS) open superstring, physical states are created by transverse bosonic and fermionic harmonic oscillators:

1. **Bosonic Oscillators:** α^i_{-n} with mode number $n \geq 1$ and transverse vector index $i = 1, \ldots, 8$, transforming in the vector representation $\mathbf{8}_v$ of SO(8).

2. **Fermionic Oscillators:** S^a_{-n} with mode number $n \geq 1$ and transverse spinor index $a = 1, \ldots, 8$, transforming in the chiral spinor representation $\mathbf{8}_s$ of SO(8).

The multi-variable partition function tracking the string oscillator mass level $q = e^{2\pi i \tau}$ (with mass-squared $\alpha' M^2 = l$) and the four SO(8) Cartan charges $x = (x_1, x_2, x_3, x_4)$ is given by:

$$\mathcal{Z}(q, x) = \prod_{n=1}^{\infty} \frac{\prod_{w \in \mathbf{8}_s} (1 + q^n e^{w \cdot x})}{\prod_{w \in \mathbf{8}_v} (1 - q^n e^{w \cdot x})} = 1 + \sum_{l=1}^{\infty} q^l \mathcal{F}_l(x). \tag{1}$$

Order by order in q, the level-l character polynomial $\mathcal{F}_l(x) = \sum_w c_w e^{w \cdot x}$ gives the exact multiplicity $c_w \in \mathbb{Z}_{\geq 0}$ of physical Fock states carrying SO(8) weight w.

1.2 Universal Massive Supersymmetry Factorization

At the massless ground state $l = 0$, the open string states form the 10D $\mathcal{N} = 1$ super-Maxwell multiplet $(\mathbf{8}_v \oplus \mathbf{8}_s)$ consisting of 8 bosons and 8 fermions. For any massive level $l \geq 1$, the fermionic zero modes satisfy the Clifford algebra $\{S^a_0, S^b_0\} = 2\delta^{ab} p^+$. Quantization of these zero modes yields the universal 256-state massive supermultiplet $\mathcal{M}_{\text{susy}}$:

$$\mathcal{M}_{\text{susy}} = (\mathbf{8}_v \oplus \mathbf{8}_s) \otimes (\mathbf{8}_v \oplus \mathbf{8}_s). \tag{2}$$

Decomposing this tensor product into irreducible SO(8) representations:

$$\text{Bosonic sector:} \quad (\mathbf{8}_v \otimes \mathbf{8}_v) \oplus (\mathbf{8}_s \otimes \mathbf{8}_s) = (\mathbf{1} \oplus \mathbf{28} \oplus \mathbf{35}_v) \oplus (\mathbf{1} \oplus \mathbf{28} \oplus \mathbf{35}_s) = \mathbf{44} \oplus \mathbf{84} = \mathbf{128}_B, \tag{3}$$

$$\text{Fermionic sector:} \quad 2 \times (\mathbf{8}_v \otimes \mathbf{8}_s) = 2 \times (\mathbf{8}_c \oplus \mathbf{56}_c) = \mathbf{128}_F. \tag{4}$$

Thus, $\mathcal{M}_{\text{susy}} = \mathbf{128}_B \oplus \mathbf{128}_F$. Spacetime supersymmetry requires that every massive level $l \geq 1$ factorizes as:

$$\mathcal{F}_l = \mathcal{M}_{\text{susy}} \otimes r^{(l)} = (128_B \oplus 128_F) \otimes (r_B^{(l)} \oplus r_F^{(l)}), \tag{5}$$

where $r^{(l)} = r_B^{(l)} \oplus r_F^{(l)}$ is the reduced *prefactor representation*. The total physical state count is:

$$N_B = N_F = 128 \times \left(\dim r_B^{(l)} + \dim r_F^{(l)} \right), \tag{6}$$

ensuring exact spacetime supersymmetry at every massive level.

1.3 The Bars Lift Proposals

- **Proposal III.B (Single Multiplicity Scheme):** Non-perturbative 11D states provide the missing components such that the cumulative sum:

$$R^{(n)} = \sum_{l=1}^{n} r^{(l)} \tag{7}$$

forms complete, non-negative irreducible representations of SO(10).

- **Proposal III.C (Higher Multiplicity Scheme):** Takes into account partition degeneracies $p(n-l)$ for longitudinal momentum modes: $R^{(n)}_{\text{III.C}} = \sum_{l=1}^{n} p(n-l) r^{(l)}$. Because Proposal III.C is analytically implied whenever Proposal III.B holds, verifying III.B is the central necessary and sufficient test.

2 Representation Theory Conventions

We use the standard orthogonal basis of highest-weight vectors $h = (h_1, \ldots, h_k)$ for the classical Lie algebras $D_4 = \mathfrak{so}(8)$, $B_4 = \mathfrak{so}(9)$, and $D_5 = \mathfrak{so}(10)$.

2.1 Lie Algebra Definitions and Weights

1. $\boldsymbol{D_4 = \mathfrak{so}(8)}$**:** Highest weights $h = (h_1, h_2, h_3, h_4)$ satisfying $h_1 \geq h_2 \geq h_3 \geq |h_4|$.
   - Vector: $(1,0,0,0) = \mathbf{8}_v$
   - Chiral Spinor: $(1/2,1/2,1/2,1/2) = \mathbf{8}_s$
   - Anti-Chiral Spinor: $(1/2,1/2,1/2,-1/2) = \mathbf{8}_c$
   - 2-form: $(1,1,0,0) = \mathbf{28}$
   - Self-Dual 4-form: $(1,1,1,1) = \mathbf{35}_s$, Anti-Self-Dual: $(1,1,1,-1) = \mathbf{35}_c$
2. $\boldsymbol{B_4 = \mathfrak{so}(9)}$**:** Highest weights $h = (h_1, h_2, h_3, h_4)$ satisfying $h_1 \geq h_2 \geq h_3 \geq h_4 \geq 0$.
   - Vector: $(1,0,0,0) = \mathbf{9}$
   - 2-form: $(1,1,0,0) = \mathbf{36}$
   - Symmetric Traceless Rank 2: $(2,0,0,0) = \mathbf{44}$
   - 3-form: $(1,1,1,0) = \mathbf{84}$
   - 4-form: $(1,1,1,1) = \mathbf{126}$
   - Majorana Spinor: $(1/2,1/2,1/2,1/2) = \mathbf{16}$
   - Vector-Spinor: $(3/2,1/2,1/2,1/2) = \mathbf{128}$
3. $\boldsymbol{D_5 = \mathfrak{so}(10)}$**:** Highest weights $H = (H_1, H_2, H_3, H_4, H_5)$ satisfying $H_1 \geq H_2 \geq H_3 \geq H_4 \geq |H_5|$.
   - Vector: $(1,0,0,0,0) = \mathbf{10}$
   - Adjoint (2-form): $(1,1,0,0,0) = \mathbf{45}$
   - Symmetric Traceless Rank 2: $(2,0,0,0,0) = \mathbf{54}$
   - 3-form: $(1,1,1,0,0) = \mathbf{120}$
   - **Chiral Self-Dual 5-form:** $(1,1,1,1,1) = \mathbf{126}$
   - **Anti-Chiral 5-form:** $(1,1,1,1,-1) = \overline{\mathbf{126}}$
   - 4-form: $(1,1,1,1,0) = \mathbf{210}$
   - Weyl Spinor: $(1/2,1/2,1/2,1/2,1/2) = \mathbf{16}$
   - Vector-Spinor: $(3/2,1/2,1/2,1/2,1/2) = \mathbf{144}$

2.2 Weyl Dimension Formulas

For $D_n = \mathfrak{so}(2n)$ and $B_n = \mathfrak{so}(2n+1)$, the dimension of an irrep with highest weight h and Weyl vector ρ is given by:

$$\dim_{D_4}(h) = \prod_{1\le i<j\le 4} \frac{(l_i - l_j)(l_i + l_j)}{(\rho_i - \rho_j)(\rho_i + \rho_j)}, \quad l = h + (3,2,1,0), \tag{8}$$

$$\dim_{B_4}(h) = \prod_{i=1}^{4} \frac{l_i}{\rho_i} \prod_{1\le i<j\le 4} \frac{(l_i - l_j)(l_i + l_j)}{(\rho_i - \rho_j)(\rho_i + \rho_j)}, \quad l = h + (3.5, 2.5, 1.5, 0.5), \tag{9}$$

$$\dim_{D_5}(H) = \prod_{1\le i<j\le 5} \frac{(l_i - l_j)(l_i + l_j)}{(\rho_i - \rho_j)(\rho_i + \rho_j)}, \quad l = H + (4,3,2,1,0). \tag{10}$$

2.3 Exact Gelfand-Tsetlin Branching Rules

1. **SO(10) → SO(9):** An irrep $H = (H_1, H_2, H_3, H_4, H_5)$ branches to all SO(9) irreps $h = (h_1, h_2, h_3, h_4)$ satisfying the interlacing inequalities:

$$H_1 \ge h_1 \ge H_2 \ge h_2 \ge H_3 \ge h_3 \ge H_4 \ge h_4 \ge |H_5|. \tag{11}$$

2. **SO(9) → SO(8):** An irrep $h = (h_1, h_2, h_3, h_4)$ branches to all SO(8) irreps $g = (g_1, g_2, g_3, g_4)$ satisfying:

$$h_1 \ge g_1 \ge h_2 \ge g_2 \ge h_3 \ge g_3 \ge h_4 \ge |g_4|. \tag{12}$$

The Critical Role of Chirality at Level 6: At Level 6, the SO(9) 4-form $(1,1,1,1) = \mathbf{126}$ appears in the string spectrum. Under the naive non-chiral lift $(1,1,1,1,0) = \mathbf{210}$, Eq. (11) gives $1 \ge h_4 \ge 0$, producing $\mathbf{126}_{\mathrm{SO}(9)} \oplus \mathbf{84}_{\mathrm{SO}(9)}$, where the spurious extra 3-form **84** forces a negative multiplicity -1 on **120**. However, lifting to the chiral self-dual 5-form $(1,1,1,1,1) = \mathbf{126}$ gives $|H_5| = 1$, which forces $h_4 = 1$ strictly, branching purely to $\mathbf{126}_{\mathrm{SO}(9)}$ without generating **84**. This completely eliminates all negative multiplicities.

3 Methodology and Group Theory Architecture in Code

The computation pipeline is implemented in pure standard Python (zero external mathematical dependencies) across a modular set of tools.

3.1 Statement of the Ab Initio Algorithm

1. **Fock Space Construction:** Construct $\mathcal{F}_l(x)$ by expanding the GS partition function (Eq. (1)) using formal multivariate polynomials on the SO(8) weight lattice.

2. **Polynomial Character Division:** Divide $\mathcal{F}_l(x)$ by $\chi_{\mathrm{susy}}(x) = \sum_{w\in\mathbf{8}_v} e^{w\cdot x} + \sum_{w\in\mathbf{8}_s} e^{w\cdot x}$ using greedy leading-monomial division in lexicographic weight order, producing the reduced quotient polynomial $Q_l(x)$.

3. SO(8) **Weyl Character Inversion:** Multiply $Q_l(x)$ by the D_4 Weyl denominator $\Delta_{D_4}(x)$ and peel off highest-weight Weyl numerators to extract the exact SO(8) irreps of $r^{(l)}$.

4. SO(8) → SO(9) **Branching Inversion:** Invert the $B_4 \to D_4$ branching rule to express $r^{(l)}$ as irreducible SO(9) representations.

5. SO(9) → SO(10) **Linear Integer Programming Lift:** For the cumulative spectrum $R_n = \sum_{l=1}^{n} r^{(l)}$, solve for the non-negative integer multiplicities $M(H) \ge 0$ under $D_5 \to B_4$ branching allowing full chiral weights $|H_5| \ge 0$.

3.2 Uniqueness of Polynomial Division via Integral Domain Property

The characters of SO(8) live in the Laurent polynomial ring $\mathcal{R} = \mathbb{Z}[x_1^{\pm 1}, x_2^{\pm 1}, x_3^{\pm 1}, x_4^{\pm 1}]$. Because the integer ring $\mathbb{Z}$ is an integral domain, $\mathcal{R}$ is also an integral domain with no non-zero zero-divisors. If $\mathcal{F}_n = Q_n \chi_{\text{susy}} = Q'_n \chi_{\text{susy}}$, subtracting gives $(Q_n - Q'_n)\chi_{\text{susy}} = 0$. Since $\chi_{\text{susy}} \neq 0$, it follows that $Q_n - Q'_n = 0 \implies Q_n = Q'_n$. Thus, the factorization is strictly unique.

3.3 Codebase Architecture and Module Reference

- `test_spectrum.py`: Core algebraic engine containing:
  - `compute_fock_levels(max_level)`: Generates the GS Fock space character polynomials $\mathcal{F}_l$.
  - `poly_div_susy(P)`: Implements lexicographic multivariate polynomial division by $(\mathbf{8}_v \oplus \mathbf{8}_s)$.
  - `decompose_so8_char(Q)`: Inverts the Weyl character formula on D_4.
  - `decompose_so8_to_so9(so8_m)`: Lifts SO(8) multiplicities to SO(9).
  - `decompose_so9_to_so10(so9_m)`: Exact backtracking solver for the SO(10) lift.
- `verify_factorization.py`: Executes the backward polynomial check $Q_n(x) \times (\mathbf{8}_v + \mathbf{8}_s) \equiv \mathcal{F}_n(x)$, confirming $\max |c_{\text{orig}} - c_{\text{recon}}| = 0$ for all 10 levels.
- `verify_so10_lifting.py`: Performs the exact character-level check by branching the lifted SO(10) irreps back to SO(9) and confirming $\Delta m(h) = 0$ for all irreps.
- `export_canonical_dataset.py`: Standalone ab initio runner generating the JSON dataset `canonical_spectrum_levels_1_to_10.json` and verification markdown report.
- `inspect_level.py`: Interactive CLI tool to inspect the full physical state accounting, SO(8)/SO(9) breakdown, and SO(10) lift for any level $l \in [1, 10]$.

3.4 Verification Summary (Levels 1 to 10)

Level n	$\dim(\mathcal{F}_n)$	r_n Prefactor	R_n SO(9) Dim	R_n SO(10) Dim	SO(10) Irreps	Result
1	16	1B + 0F	1B + 0F	1B + 0F	1	**PASSE**
2	144	9B + 0F	10B + 0F	10B + 0F	1	**PASSE**
3	960	44B + 16F	54B + 16F	54B + 16F	2	**PASSE**
4	5,264	201B + 128F	255B + 144F	255B + 144F	3	**PASSE**
5	25,056	846B + 720F	1,101B + 864F	1,101B + 864F	6	**PASSE**
6	106,944	3,468B + 3,216F	4,569B + 4,080F	4,569B + 4,080F	11	**PASSE**
7	418,176	13,240B + 12,896F	17,809B + 16,976F	17,809B + 16,976F	18	**PASSE**
8	1,520,784	47,817B + 47,232F	65,626B + 64,208F	65,626B + 64,208F	29	**PASSE**
9	5,201,232	162,917B + 162,160F	228,543B + 226,368F	228,543B + 226,368F	39	**PASSE**
10	16,871,648	527,806B + 526,672F	756,349B + 753,040F	756,349B + 753,040F	53	**PASSE**

Table 1: Exact dimension matching and SO(10) lifting verification for Levels 1 through 10.

A Detailed Level-by-Level Verification Data

This appendix contains the complete representation-theoretic breakdown for each mass level $n = 1, \ldots, 10$, including single-level prefactors r_n, cumulative SO(9) spectra $R_n = \sum_{l=1}^{n} r_l$, lifted SO(10) supermultiplets, and term-by-term branching verification tables.

A.1 Level $n = 1$ Representation Data

Single-Level Prefactor r_n (in SO(9)):

- **Bosons:** $1 \times$ **1** `(0, 0, 0, 0)`
- **Fermions:** None

Cumulative Spectrum $R_n = \sum_{l=1}^{n} r_l$ (in SO(9)):

- **Bosons:** $1 \times$ **1** `(0, 0, 0, 0)`
- **Fermions:** None

SO(10) **Supermultiplet Decomposition:**

- **Bosons:** $1 \times$ **1** `(0, 0, 0, 0, 0)`
- **Fermions:** None

SO(9) **Term-by-Term Branching Reconstruction Verification:**

Type	SO(9) Weight	Dim	Target Mult (R_n)	Recon Mult (from SO(10))	Difference
B	`(0, 0, 0, 0)`	**1**	1	1	0

Table 2: Level 1 term-by-term SO(9) branching reconstruction verification (exact difference = 0).

A.2 Level $n = 2$ Representation Data

Single-Level Prefactor r_n (in SO(9)):

- **Bosons:** $1 \times$ **9** `(1, 0, 0, 0)`
- **Fermions:** None

Cumulative Spectrum $R_n = \sum_{l=1}^{n} r_l$ (in SO(9)):

- **Bosons:** $1 \times$ **1** `(0, 0, 0, 0)`, $1 \times$ **9** `(1, 0, 0, 0)`
- **Fermions:** None

SO(10) **Supermultiplet Decomposition:**

- **Bosons:** $1 \times$ **10** `(1, 0, 0, 0, 0)`
- **Fermions:** None

SO(9) **Term-by-Term Branching Reconstruction Verification:**

Type	SO(9) Weight	Dim	Target Mult (R_n)	Recon Mult (from SO(10))	Difference
B	(1, 0, 0, 0)	**9**	1	1	0
B	(0, 0, 0, 0)	**1**	1	1	0

Table 3: Level 2 term-by-term SO(9) branching reconstruction verification (exact difference = 0).

A.3 Level $n = 3$ Representation Data

Single-Level Prefactor r_n (in SO(9)):

- **Bosons:** $1 \times$ **44** (2, 0, 0, 0)
- **Fermions:** $1 \times$ **16** (0.5, 0.5, 0.5, 0.5)

Cumulative Spectrum $R_n = \sum_{l=1}^{n} r_l$ (in SO(9)):

- **Bosons:** $1 \times$ **1** (0, 0, 0, 0), $1 \times$ **9** (1, 0, 0, 0), $1 \times$ **44** (2, 0, 0, 0)
- **Fermions:** $1 \times$ **16** (0.5, 0.5, 0.5, 0.5)

SO(10) **Supermultiplet Decomposition:**

- **Bosons:** $1 \times$ **54** (2, 0, 0, 0, 0)
- **Fermions:** $1 \times$ **16** (0.5, 0.5, 0.5, 0.5, 0.5)

SO(9) **Term-by-Term Branching Reconstruction Verification:**

Type	SO(9) Weight	Dim	Target Mult (R_n)	Recon Mult (from SO(10))	Difference
B	(2, 0, 0, 0)	**44**	1	1	0
B	(1, 0, 0, 0)	**9**	1	1	0
B	(0, 0, 0, 0)	**1**	1	1	0
F	(0.5, 0.5, 0.5, 0.5)	**16**	1	1	0

Table 4: Level 3 term-by-term SO(9) branching reconstruction verification (exact difference = 0).

A.4 Level $n = 4$ Representation Data

Single-Level Prefactor r_n (in SO(9)):

- **Bosons:** $1 \times$ **156** (3, 0, 0, 0), $1 \times$ **36** (1, 1, 0, 0), $1 \times$ **9** (1, 0, 0, 0)
- **Fermions:** $1 \times$ **128** (1.5, 0.5, 0.5, 0.5)

Cumulative Spectrum $R_n = \sum_{l=1}^{n} r_l$ (in SO(9)):

- **Bosons:** $1 \times$ **1** (0, 0, 0, 0), $2 \times$ **9** (1, 0, 0, 0), $1 \times$ **44** (2, 0, 0, 0), $1 \times$ **156** (3, 0, 0, 0), $1 \times$ **36** (1, 1, 0, 0)
- **Fermions:** $1 \times$ **16** (0.5, 0.5, 0.5, 0.5), $1 \times$ **128** (1.5, 0.5, 0.5, 0.5)

SO(10) **Supermultiplet Decomposition:**

- **Bosons:** $1 \times$ **45** (1, 1, 0, 0, 0), $1 \times$ **210** (3, 0, 0, 0, 0)
- **Fermions:** $1 \times$ **144** (1.5, 0.5, 0.5, 0.5, 0.5)

SO(9) **Term-by-Term Branching Reconstruction Verification:**

Type	SO(9) Weight	Dim	Target Mult (R_n)	Recon Mult (from SO(10))	Difference
B	(3, 0, 0, 0)	**156**	1	1	0
B	(2, 0, 0, 0)	**44**	1	1	0
B	(1, 1, 0, 0)	**36**	1	1	0
B	(1, 0, 0, 0)	**9**	2	2	0
B	(0, 0, 0, 0)	**1**	1	1	0
F	(1.5, 0.5, 0.5, 0.5)	**128**	1	1	0
F	(0.5, 0.5, 0.5, 0.5)	**16**	1	1	0

Table 5: Level 4 term-by-term SO(9) branching reconstruction verification (exact difference = 0).

A.5 Level $n = 5$ Representation Data

Single-Level Prefactor r_n (in SO(9)):

- **Bosons:** $1 \times$ **450** (4, 0, 0, 0), $1 \times$ **231** (2, 1, 0, 0), $1 \times$ **44** (2, 0, 0, 0), $1 \times$ **84** (1, 1, 1, 0), $1 \times$ **36** (1, 1, 0, 0), $1 \times$ **1** (0, 0, 0, 0)
- **Fermions:** $1 \times$ **576** (2.5, 0.5, 0.5, 0.5), $1 \times$ **128** (1.5, 0.5, 0.5, 0.5), $1 \times$ **16** (0.5, 0.5, 0.5, 0.5)

Cumulative Spectrum $R_n = \sum_{l=1}^{n} r_l$ (in SO(9)):

- **Bosons:** $2 \times$ **1** (0, 0, 0, 0), $2 \times$ **9** (1, 0, 0, 0), $2 \times$ **44** (2, 0, 0, 0), $1 \times$ **156** (3, 0, 0, 0), $2 \times$ **36** (1, 1, 0, 0), $1 \times$ **450** (4, 0, 0, 0), $1 \times$ **231** (2, 1, 0, 0), $1 \times$ **84** (1, 1, 1, 0)
- **Fermions:** $2 \times$ **16** (0.5, 0.5, 0.5, 0.5), $2 \times$ **128** (1.5, 0.5, 0.5, 0.5), $1 \times$ **576** (2.5, 0.5, 0.5, 0.5)

SO(10) **Supermultiplet Decomposition:**

- **Bosons:** $1 \times$ **1** (0, 0, 0, 0, 0), $1 \times$ **120** (1, 1, 1, 0, 0), $1 \times$ **320** (2, 1, 0, 0, 0), $1 \times$ **660** (4, 0, 0, 0, 0)
- **Fermions:** $1 \times$ **144** (1.5, 0.5, 0.5, 0.5, 0.5), $1 \times$ **720** (2.5, 0.5, 0.5, 0.5, 0.5)

SO(9) **Term-by-Term Branching Reconstruction Verification:**

Type	SO(9) Weight	Dim	Target Mult (R_n)	Recon Mult (from SO(10))	Difference
B	(4, 0, 0, 0)	**450**	1	1	0
B	(2, 1, 0, 0)	**231**	1	1	0
B	(3, 0, 0, 0)	**156**	1	1	0

Type	SO(9) Weight	Dim	Target Mult (R_n)	Recon Mult (from SO(10))	Difference
B	(1, 1, 1, 0)	**84**	1	1	0
B	(2, 0, 0, 0)	**44**	2	2	0
B	(1, 1, 0, 0)	**36**	2	2	0
B	(1, 0, 0, 0)	**9**	2	2	0
B	(0, 0, 0, 0)	**1**	2	2	0
F	(2.5, 0.5, 0.5, 0.5)	**576**	1	1	0
F	(1.5, 0.5, 0.5, 0.5)	**128**	2	2	0
F	(0.5, 0.5, 0.5, 0.5)	**16**	2	2	0

Table 6: Level 5 term-by-term SO(9) branching reconstruction verification (exact difference = 0).

A.6 Level $n = 6$ Representation Data

Single-Level Prefactor r_n (in SO(9)):

- **Bosons:** $1\times$**1122** (5, 0, 0, 0), $1\times$**910** (3, 1, 0, 0), $1\times$**156** (3, 0, 0, 0), $1\times$**594** (2, 1, 1, 0), $2\times$**231** (2, 1, 0, 0), $1\times$**44** (2, 0, 0, 0), $1\times$**126** (1, 1, 1, 1), $1\times$**36** (1, 1, 0, 0), $2\times$**9** (1, 0, 0, 0)
- **Fermions:** $1\times$**1920** (3.5, 0.5, 0.5, 0.5), $1\times$**576** (2.5, 0.5, 0.5, 0.5), $1\times$**432** (1.5, 1.5, 0.5, 0.5), $2\times$**128** (1.5, 0.5, 0.5, 0.5), $2\times$**16** (0.5, 0.5, 0.5, 0.5)

Cumulative Spectrum $R_n = \sum_{l=1}^{n} r_l$ (in SO(9)):

- **Bosons:** $2\times$**1** (0, 0, 0, 0), $4\times$**9** (1, 0, 0, 0), $3\times$**44** (2, 0, 0, 0), $2\times$**156** (3, 0, 0, 0), $3\times$**36** (1, 1, 0, 0), $1\times$**450** (4, 0, 0, 0), $3\times$**231** (2, 1, 0, 0), $1\times$**84** (1, 1, 1, 0), $1\times$**1122** (5, 0, 0, 0), $1\times$**910** (3, 1, 0, 0), $1\times$**594** (2, 1, 1, 0), $1\times$**126** (1, 1, 1, 1)
- **Fermions:** $4\times$**16** (0.5, 0.5, 0.5, 0.5), $4\times$**128** (1.5, 0.5, 0.5, 0.5), $2\times$**576** (2.5, 0.5, 0.5, 0.5), $1\times$**1920** (3.5, 0.5, 0.5, 0.5), $1\times$**432** (1.5, 1.5, 0.5, 0.5)

SO(10) **Supermultiplet Decomposition:**

- **Bosons:** $1\times$**10** (1, 0, 0, 0, 0), $1\times$**126** (1, 1, 1, 1, 1), $1\times$**320** (2, 1, 0, 0, 0), $1\times$**945** (2, 1, 1, 0, 0), $1\times$**1386** (3, 1, 0, 0, 0), $1\times$**1782** (5, 0, 0, 0, 0)
- **Fermions:** $1\times$**16** (0.5, 0.5, 0.5, 0.5, 0.5), $1\times$**144** (1.5, 0.5, 0.5, 0.5, 0.5), $1\times$**560** (1.5, 1.5, 0.5, 0.5, 0.5), $1\times$**720** (2.5, 0.5, 0.5, 0.5, 0.5), $1\times$**2640** (3.5, 0.5, 0.5, 0.5, 0.5)

SO(9) **Term-by-Term Branching Reconstruction Verification:**

Type	SO(9) Weight	Dim	Target Mult (R_n)	Recon Mult (from SO(10))	Difference
B	(5, 0, 0, 0)	**1122**	1	1	0
B	(3, 1, 0, 0)	**910**	1	1	0
B	(2, 1, 1, 0)	**594**	1	1	0
B	(4, 0, 0, 0)	**450**	1	1	0
B	(2, 1, 0, 0)	**231**	3	3	0

Type	SO(9) Weight	Dim	Target Mult (R_n)	Recon Mult (from SO(10))	Difference
B	(3, 0, 0, 0)	**156**	2	2	0
B	(1, 1, 1, 1)	**126**	1	1	0
B	(1, 1, 1, 0)	**84**	1	1	0
B	(2, 0, 0, 0)	**44**	3	3	0
B	(1, 1, 0, 0)	**36**	3	3	0
B	(1, 0, 0, 0)	**9**	4	4	0
B	(0, 0, 0, 0)	**1**	2	2	0
F	(3.5, 0.5, 0.5, 0.5)	**1920**	1	1	0
F	(2.5, 0.5, 0.5, 0.5)	**576**	2	2	0
F	(1.5, 1.5, 0.5, 0.5)	**432**	1	1	0
F	(1.5, 0.5, 0.5, 0.5)	**128**	4	4	0
F	(0.5, 0.5, 0.5, 0.5)	**16**	4	4	0

Table 7: Level 6 term-by-term SO(9) branching reconstruction verification (exact difference = 0).

A.7 Level $n = 7$ Representation Data

Single-Level Prefactor r_n (in SO(9)):

- **Bosons:** $1 \times$ **2508** (6, 0, 0, 0), $1 \times$ **2772** (4, 1, 0, 0), $1 \times$ **450** (4, 0, 0, 0), $1 \times$ **2457** (3, 1, 1, 0), $2 \times$ **910** (3, 1, 0, 0), $1 \times$ **156** (3, 0, 0, 0), $1 \times$ **495** (2, 2, 0, 0), $1 \times$ **924** (2, 1, 1, 1), $1 \times$ **594** (2, 1, 1, 0), $2 \times$ **231** (2, 1, 0, 0), $3 \times$ **44** (2, 0, 0, 0), $1 \times$ **126** (1, 1, 1, 1), $3 \times$ **84** (1, 1, 1, 0), $2 \times$ **36** (1, 1, 0, 0), $2 \times$ **9** (1, 0, 0, 0), $2 \times$ **1** (0, 0, 0, 0)
- **Fermions:** $1 \times$ **5280** (4.5, 0.5, 0.5, 0.5), $1 \times$ **1920** (3.5, 0.5, 0.5, 0.5), $1 \times$ **2560** (2.5, 1.5, 0.5, 0.5), $3 \times$ **576** (2.5, 0.5, 0.5, 0.5), $2 \times$ **432** (1.5, 1.5, 0.5, 0.5), $4 \times$ **128** (1.5, 0.5, 0.5, 0.5), $2 \times$ **16** (0.5, 0.5, 0.5, 0.5)

Cumulative Spectrum $R_n = \sum_{l=1}^{n} r_l$ (in SO(9)):

- **Bosons:** $4 \times$ **1** (0, 0, 0, 0), $6 \times$ **9** (1, 0, 0, 0), $6 \times$ **44** (2, 0, 0, 0), $3 \times$ **156** (3, 0, 0, 0), $5 \times$ **36** (1, 1, 0, 0), $2 \times$ **450** (4, 0, 0, 0), $5 \times$ **231** (2, 1, 0, 0), $4 \times$ **84** (1, 1, 1, 0), $1 \times$ **1122** (5, 0, 0, 0), $3 \times$ **910** (3, 1, 0, 0), $2 \times$ **594** (2, 1, 1, 0), $2 \times$ **126** (1, 1, 1, 1), $1 \times$ **2508** (6, 0, 0, 0), $1 \times$ **2772** (4, 1, 0, 0), $1 \times$ **2457** (3, 1, 1, 0), $1 \times$ **495** (2, 2, 0, 0), $1 \times$ **924** (2, 1, 1, 1)
- **Fermions:** $6 \times$ **16** (0.5, 0.5, 0.5, 0.5), $8 \times$ **128** (1.5, 0.5, 0.5, 0.5), $5 \times$ **576** (2.5, 0.5, 0.5, 0.5), $2 \times$ **1920** (3.5, 0.5, 0.5, 0.5), $3 \times$ **432** (1.5, 1.5, 0.5, 0.5), $1 \times$ **5280** (4.5, 0.5, 0.5, 0.5), $1 \times$ **2560** (2.5, 1.5, 0.5, 0.5)

SO(10) **Supermultiplet Decomposition:**

- **Bosons:** $1 \times$ **10** (1, 0, 0, 0, 0), $1 \times$ **120** (1, 1, 1, 0, 0), $1 \times$ **210** (1, 1, 1, 1, 0), $2 \times$ **54** (2, 0, 0, 0, 0), $1 \times$ **945** (2, 1, 1, 0, 0), $1 \times$ **1050** (2, 1, 1, 1, 1), $1 \times$ **770** (2, 2, 0, 0, 0), $1 \times$ **1386** (3, 1, 0, 0, 0), $1 \times$ **4312** (3, 1, 1, 0, 0), $1 \times$ **4608** (4, 1, 0, 0, 0), $1 \times$ **4290** (6, 0, 0, 0, 0)
- **Fermions:** $1 \times$ **16** (0.5, 0.5, 0.5, 0.5, 0.5), $1 \times$ **144** (1.5, 0.5, 0.5, 0.5, 0.5), $2 \times$ **560** (1.5, 1.5, 0.5, 0.5, 0.5), $2 \times$ **720** (2.5, 0.5, 0.5, 0.5, 0.5), $1 \times$ **3696** (2.5, 1.5, 0.5, 0.5, 0.5), $1 \times$ **2640** (3.5, 0.5, 0.5, 0.5, 0.5), $1 \times$ **7920** (4.5, 0.5, 0.5, 0.5, 0.5)

SO(9) **Term-by-Term Branching Reconstruction Verification:**

Type	SO(9) Weight	Dim	Target Mult (R_n)	Recon Mult (from SO(10))	Difference
B	(4, 1, 0, 0)	**2772**	1	1	0
B	(6, 0, 0, 0)	**2508**	1	1	0
B	(3, 1, 1, 0)	**2457**	1	1	0
B	(5, 0, 0, 0)	**1122**	1	1	0
B	(2, 1, 1, 1)	**924**	1	1	0
B	(3, 1, 0, 0)	**910**	3	3	0
B	(2, 1, 1, 0)	**594**	2	2	0
B	(2, 2, 0, 0)	**495**	1	1	0
B	(4, 0, 0, 0)	**450**	2	2	0
B	(2, 1, 0, 0)	**231**	5	5	0
B	(3, 0, 0, 0)	**156**	3	3	0
B	(1, 1, 1, 1)	**126**	2	2	0
B	(1, 1, 1, 0)	**84**	4	4	0
B	(2, 0, 0, 0)	**44**	6	6	0
B	(1, 1, 0, 0)	**36**	5	5	0
B	(1, 0, 0, 0)	**9**	6	6	0
B	(0, 0, 0, 0)	**1**	4	4	0
F	(4.5, 0.5, 0.5, 0.5)	**5280**	1	1	0
F	(2.5, 1.5, 0.5, 0.5)	**2560**	1	1	0
F	(3.5, 0.5, 0.5, 0.5)	**1920**	2	2	0
F	(2.5, 0.5, 0.5, 0.5)	**576**	5	5	0
F	(1.5, 1.5, 0.5, 0.5)	**432**	3	3	0
F	(1.5, 0.5, 0.5, 0.5)	**128**	8	8	0
F	(0.5, 0.5, 0.5, 0.5)	**16**	6	6	0

Table 8: Level 7 term-by-term SO(9) branching reconstruction verification (exact difference = 0).

A.8 Level $n = 8$ Representation Data

Single-Level Prefactor r_n (in SO(9)):

- **Bosons:** $1 \times$ **5148** (7, 0, 0, 0), $1 \times$ **7140** (5, 1, 0, 0), $1 \times$ **1122** (5, 0, 0, 0), $1 \times$ **7700** (4, 1, 1, 0), $2 \times$ **2772** (4, 1, 0, 0), $1 \times$ **450** (4, 0, 0, 0), $1 \times$ **2574** (3, 2, 0, 0), $1 \times$ **3900** (3, 1, 1, 1), $1 \times$ **2457** (3, 1, 1, 0), $3 \times$ **910** (3, 1, 0, 0), $4 \times$ **156** (3, 0, 0, 0), $1 \times$ **1650** (2, 2, 1, 0), $1 \times$ **495** (2, 2, 0, 0), $2 \times$ **924** (2, 1, 1, 1), $4 \times$ **594** (2, 1, 1, 0), $5 \times$ **231** (2, 1, 0, 0), $3 \times$ **44** (2, 0, 0, 0), $3 \times$ **126** (1, 1, 1, 1), $2 \times$ **84** (1, 1, 1, 0), $5 \times$ **36** (1, 1, 0, 0), $5 \times$ **9** (1, 0, 0, 0), $1 \times$ **1** (0, 0, 0, 0)
- **Fermions:** $1 \times$ **12672** (5.5, 0.5, 0.5, 0.5), $1 \times$ **5280** (4.5, 0.5, 0.5, 0.5), $1 \times$ **9504** (3.5, 1.5, 0.5, 0.5), $3 \times$ **1920** (3.5, 0.5, 0.5, 0.5), $3 \times$ **2560** (2.5, 1.5, 0.5, 0.5), $5 \times$ **576** (2.5, 0.5, 0.5, 0.5), $1 \times$ **768** (1.5, 1.5, 1.5, 0.5), $4 \times$ **432** (1.5, 1.5, 0.5, 0.5), $7 \times$ **128** (1.5, 0.5, 0.5, 0.5), $4 \times$ **16** (0.5, 0.5, 0.5, 0.5)

Cumulative Spectrum $R_n = \sum_{l=1}^{n} r_l$ (in SO(9)):

- **Bosons:** $5 \times$ **1** (0, 0, 0, 0), $11 \times$ **9** (1, 0, 0, 0), $9 \times$ **44** (2, 0, 0, 0), $7 \times$ **156** (3, 0, 0, 0), $10 \times$ **36** (1, 1, 0, 0), $3 \times$ **450** (4, 0, 0, 0), $10 \times$ **231** (2, 1, 0, 0),

$6 \times$ **84** (1, 1, 1, 0), $2 \times$ **1122** (5, 0, 0, 0), $6 \times$ **910** (3, 1, 0, 0), $6 \times$ **594** (2, 1, 1, 0), $5 \times$ **126** (1, 1, 1, 1), $1 \times$ **2508** (6, 0, 0, 0), $3 \times$ **2772** (4, 1, 0, 0), $2 \times$ **2457** (3, 1, 1, 0), $2 \times$ **495** (2, 2, 0, 0), $3 \times$ **924** (2, 1, 1, 1), $1 \times$ **5148** (7, 0, 0, 0), $1 \times$ **7140** (5, 1, 0, 0), $1 \times$ **7700** (4, 1, 1, 0), $1 \times$ **2574** (3, 2, 0, 0), $1 \times$ **3900** (3, 1, 1, 1), $1 \times$ **1650** (2, 2, 1, 0)

- **Fermions:** $10 \times$ **16** (0.5, 0.5, 0.5, 0.5), $15 \times$ **128** (1.5, 0.5, 0.5, 0.5), $10 \times$ **576** (2.5, 0.5, 0.5, 0.5), $5 \times$ **1920** (3.5, 0.5, 0.5, 0.5), $7 \times$ **432** (1.5, 1.5, 0.5, 0.5), $2 \times$ **5280** (4.5, 0.5, 0.5, 0.5), $4 \times$ **2560** (2.5, 1.5, 0.5, 0.5), $1 \times$ **12672** (5.5, 0.5, 0.5, 0.5), $1 \times$ **9504** (3.5, 1.5, 0.5, 0.5), $1 \times$ **768** (1.5, 1.5, 1.5, 0.5)

SO(10) **Supermultiplet Decomposition:**

- **Bosons:** $1 \times$ **10** (1, 0, 0, 0, 0), $2 \times$ **45** (1, 1, 0, 0, 0), $1 \times$ **210** (1, 1, 1, 1, 0), $1 \times$ **126** (1, 1, 1, 1, 1), $1 \times$ **54** (2, 0, 0, 0, 0), $1 \times$ **320** (2, 1, 0, 0, 0), $2 \times$ **945** (2, 1, 1, 0, 0), $1 \times$ **1728** (2, 1, 1, 1, 0), $1 \times$ **1050** (2, 1, 1, 1, 1), $1 \times$ **2970** (2, 2, 1, 0, 0), $2 \times$ **210** (3, 0, 0, 0, 0), $1 \times$ **1386** (3, 1, 0, 0, 0), $1 \times$ **4312** (3, 1, 1, 0, 0), $1 \times$ **4950** (3, 1, 1, 1, 1), $1 \times$ **4410** (3, 2, 0, 0, 0), $1 \times$ **4608** (4, 1, 0, 0, 0), $1 \times$ **14784** (4, 1, 1, 0, 0), $1 \times$ **12870** (5, 1, 0, 0, 0), $1 \times$ **9438** (7, 0, 0, 0, 0)
- **Fermions:** $1 \times$ **16** (0.5, 0.5, 0.5, 0.5, 0.5), $3 \times$ **144** (1.5, 0.5, 0.5, 0.5, 0.5), $2 \times$ **560** (1.5, 1.5, 0.5, 0.5, 0.5), $1 \times$ **1200** (1.5, 1.5, 1.5, 0.5, 0.5), $2 \times$ **720** (2.5, 0.5, 0.5, 0.5, 0.5), $3 \times$ **3696** (2.5, 1.5, 0.5, 0.5, 0.5), $2 \times$ **2640** (3.5, 0.5, 0.5, 0.5, 0.5), $1 \times$ **15120** (3.5, 1.5, 0.5, 0.5, 0.5), $1 \times$ **7920** (4.5, 0.5, 0.5, 0.5, 0.5), $1 \times$ **20592** (5.5, 0.5, 0.5, 0.5, 0.5)

SO(9) **Term-by-Term Branching Reconstruction Verification:**

Type	SO(9) Weight	Dim	Target Mult (R_n)	Recon Mult (from SO(10))	Difference
B	(4, 1, 1, 0)	**7700**	1	1	0
B	(5, 1, 0, 0)	**7140**	1	1	0
B	(7, 0, 0, 0)	**5148**	1	1	0
B	(3, 1, 1, 1)	**3900**	1	1	0
B	(4, 1, 0, 0)	**2772**	3	3	0
B	(3, 2, 0, 0)	**2574**	1	1	0
B	(6, 0, 0, 0)	**2508**	1	1	0
B	(3, 1, 1, 0)	**2457**	2	2	0
B	(2, 2, 1, 0)	**1650**	1	1	0
B	(5, 0, 0, 0)	**1122**	2	2	0
B	(2, 1, 1, 1)	**924**	3	3	0
B	(3, 1, 0, 0)	**910**	6	6	0
B	(2, 1, 1, 0)	**594**	6	6	0
B	(2, 2, 0, 0)	**495**	2	2	0
B	(4, 0, 0, 0)	**450**	3	3	0
B	(2, 1, 0, 0)	**231**	10	10	0
B	(3, 0, 0, 0)	**156**	7	7	0
B	(1, 1, 1, 1)	**126**	5	5	0
B	(1, 1, 1, 0)	**84**	6	6	0
B	(2, 0, 0, 0)	**44**	9	9	0

Type	SO(9) Weight	Dim	Target Mult (R_n)	Recon Mult (from SO(10))	Difference
B	(1, 1, 0, 0)	**36**	10	10	0
B	(1, 0, 0, 0)	**9**	11	11	0
B	(0, 0, 0, 0)	**1**	5	5	0
F	(5.5, 0.5, 0.5, 0.5)	**12672**	1	1	0
F	(3.5, 1.5, 0.5, 0.5)	**9504**	1	1	0
F	(4.5, 0.5, 0.5, 0.5)	**5280**	2	2	0
F	(2.5, 1.5, 0.5, 0.5)	**2560**	4	4	0
F	(3.5, 0.5, 0.5, 0.5)	**1920**	5	5	0
F	(1.5, 1.5, 1.5, 0.5)	**768**	1	1	0
F	(2.5, 0.5, 0.5, 0.5)	**576**	10	10	0
F	(1.5, 1.5, 0.5, 0.5)	**432**	7	7	0
F	(1.5, 0.5, 0.5, 0.5)	**128**	15	15	0
F	(0.5, 0.5, 0.5, 0.5)	**16**	10	10	0

Table 9: Level 8 term-by-term SO(9) branching reconstruction verification (exact difference = 0).

A.9 Level $n = 9$ Representation Data

Single-Level Prefactor r_n (in SO(9)):

- **Bosons:** $1 \times$ **9867** (8, 0, 0, 0), $1 \times$ **16302** (6, 1, 0, 0), $1 \times$ **2508** (6, 0, 0, 0), $1 \times$ **20196** (5, 1, 1, 0), $2 \times$ **7140** (5, 1, 0, 0), $1 \times$ **1122** (5, 0, 0, 0), $1 \times$ **8748** (4, 2, 0, 0), $1 \times$ **12375** (4, 1, 1, 1), $1 \times$ **7700** (4, 1, 1, 0), $3 \times$ **2772** (4, 1, 0, 0), $4 \times$ **450** (4, 0, 0, 0), $1 \times$ **9009** (3, 2, 1, 0), $2 \times$ **2574** (3, 2, 0, 0), $2 \times$ **3900** (3, 1, 1, 1), $5 \times$ **2457** (3, 1, 1, 0), $6 \times$ **910** (3, 1, 0, 0), $4 \times$ **156** (3, 0, 0, 0), $1 \times$ **2772** (2, 2, 1, 1), $1 \times$ **1650** (2, 2, 1, 0), $4 \times$ **495** (2, 2, 0, 0), $6 \times$ **924** (2, 1, 1, 1), $6 \times$ **594** (2, 1, 1, 0), $9 \times$ **231** (2, 1, 0, 0), $9 \times$ **44** (2, 0, 0, 0), $4 \times$ **126** (1, 1, 1, 1), $7 \times$ **84** (1, 1, 1, 0), $7 \times$ **36** (1, 1, 0, 0), $5 \times$ **9** (1, 0, 0, 0), $3 \times$ **1** (0, 0, 0, 0)
- **Fermions:** $1 \times$ **27456** (6.5, 0.5, 0.5, 0.5), $1 \times$ **12672** (5.5, 0.5, 0.5, 0.5), $1 \times$ **27648** (4.5, 1.5, 0.5, 0.5), $3 \times$ **5280** (4.5, 0.5, 0.5, 0.5), $3 \times$ **9504** (3.5, 1.5, 0.5, 0.5), $6 \times$ **1920** (3.5, 0.5, 0.5, 0.5), $1 \times$ **4928** (2.5, 2.5, 0.5, 0.5), $1 \times$ **5040** (2.5, 1.5, 1.5, 0.5), $6 \times$ **2560** (2.5, 1.5, 0.5, 0.5), $10 \times$ **576** (2.5, 0.5, 0.5, 0.5), $3 \times$ **768** (1.5, 1.5, 1.5, 0.5), $8 \times$ **432** (1.5, 1.5, 0.5, 0.5), $12 \times$ **128** (1.5, 0.5, 0.5, 0.5), $8 \times$ **16** (0.5, 0.5, 0.5, 0.5)

Cumulative Spectrum $R_n = \sum_{l=1}^{n} r_l$ (in SO(9)):

- **Bosons:** $8 \times$ **1** (0, 0, 0, 0), $16 \times$ **9** (1, 0, 0, 0), $18 \times$ **44** (2, 0, 0, 0), $11 \times$ **156** (3, 0, 0, 0), $17 \times$ **36** (1, 1, 0, 0), $7 \times$ **450** (4, 0, 0, 0), $19 \times$ **231** (2, 1, 0, 0), $13 \times$ **84** (1, 1, 1, 0), $3 \times$ **1122** (5, 0, 0, 0), $12 \times$ **910** (3, 1, 0, 0), $12 \times$ **594** (2, 1, 1, 0), $9 \times$ **126** (1, 1, 1, 1), $2 \times$ **2508** (6, 0, 0, 0), $6 \times$ **2772** (4, 1, 0, 0), $7 \times$ **2457** (3, 1, 1, 0), $6 \times$ **495** (2, 2, 0, 0), $9 \times$ **924** (2, 1, 1, 1), $1 \times$ **5148** (7, 0, 0, 0), $3 \times$ **7140** (5, 1, 0, 0), $2 \times$ **7700** (4, 1, 1, 0), $3 \times$ **2574** (3, 2, 0, 0), $3 \times$ **3900** (3, 1, 1, 1), $2 \times$ **1650** (2, 2, 1, 0), $1 \times$ **9867** (8, 0, 0, 0), $1 \times$ **16302** (6, 1, 0, 0), $1 \times$ **20196** (5, 1, 1, 0), $1 \times$ **8748** (4, 2, 0, 0), $1 \times$ **12375** (4, 1, 1, 1), $1 \times$ **9009** (3, 2, 1, 0), $1 \times$ **2772** (2, 2, 1, 1)
- **Fermions:** $18 \times$ **16** (0.5, 0.5, 0.5, 0.5), $27 \times$ **128** (1.5, 0.5, 0.5, 0.5), $20 \times$ **576** (2.5, 0.5, 0.5, 0.5), $11 \times$ **1920** (3.5, 0.5, 0.5, 0.5), $15 \times$ **432** (1.5, 1.5, 0.5,

0.5), $5 \times$ **5280** (4.5, 0.5, 0.5, 0.5), $10 \times$ **2560** (2.5, 1.5, 0.5, 0.5), $2 \times$ **12672** (5.5, 0.5, 0.5, 0.5), $4 \times$ **9504** (3.5, 1.5, 0.5, 0.5), $4 \times$ **768** (1.5, 1.5, 1.5, 0.5), $1 \times$ **27456** (6.5, 0.5, 0.5, 0.5), $1 \times$ **27648** (4.5, 1.5, 0.5, 0.5), $1 \times$ **4928** (2.5, 2.5, 0.5, 0.5), $1 \times$ **5040** (2.5, 1.5, 1.5, 0.5)

SO(10) **Supermultiplet Decomposition:**

- **Bosons:** $2\times$**45** (1, 1, 0, 0, 0), $2\times$**120** (1, 1, 1, 0, 0), $1\times$**210** (1, 1, 1, 1, 0), $2\times$**54** (2, 0, 0, 0, 0), $3\times$**320** (2, 1, 0, 0, 0), $1\times$**945** (2, 1, 1, 0, 0), $3\times$**1728** (2, 1, 1, 1, 0), $2\times$**1050** (2, 1, 1, 1, 1), $2\times$**770** (2, 2, 0, 0, 0), $1\times$**2970** (2, 2, 1, 0, 0), $1\times$**3696** (2, 2, 1, 1, 1), $3\times$**210** (3, 0, 0, 0, 0), $4\times$**4312** (3, 1, 1, 0, 0), $2\times$**4950** (3, 1, 1, 1, 1), $1\times$**4410** (3, 2, 0, 0, 0), $1\times$**17920** (3, 2, 1, 0, 0), $2\times$**660** (4, 0, 0, 0, 0), $1\times$**4608** (4, 1, 0, 0, 0), $1\times$**14784** (4, 1, 1, 0, 0), $1\times$**17325** (4, 1, 1, 1, 1), $1\times$**16380** (4, 2, 0, 0, 0), $1\times$**12870** (5, 1, 0, 0, 0), $1\times$**42120** (5, 1, 1, 0, 0), $1\times$**31680** (6, 1, 0, 0, 0), $1\times$**19305** (8, 0, 0, 0, 0)
- **Fermions:** $2\times$**16** (0.5, 0.5, 0.5, 0.5, 0.5), $5\times$**144** (1.5, 0.5, 0.5, 0.5, 0.5), $3\times$**560** (1.5, 1.5, 0.5, 0.5, 0.5), $3\times$**1200** (1.5, 1.5, 1.5, 0.5, 0.5), $4\times$**720** (2.5, 0.5, 0.5, 0.5, 0.5), $4\times$**3696** (2.5, 1.5, 0.5, 0.5, 0.5), $1\times$**8800** (2.5, 1.5, 1.5, 0.5, 0.5), $1\times$**8064** (2.5, 2.5, 0.5, 0.5, 0.5), $3\times$**2640** (3.5, 0.5, 0.5, 0.5, 0.5), $3\times$**15120** (3.5, 1.5, 0.5, 0.5, 0.5), $2\times$**7920** (4.5, 0.5, 0.5, 0.5, 0.5), $1\times$**48048** (4.5, 1.5, 0.5, 0.5, 0.5), $1\times$**20592** (5.5, 0.5, 0.5, 0.5, 0.5), $1\times$**48048** (6.5, 0.5, 0.5, 0.5, 0.5)

SO(9) **Term-by-Term Branching Reconstruction Verification:**

Type	SO(9) Weight	Dim	Target Mult (R_n)	Recon Mult (from SO(10))	Difference
B	(5, 1, 1, 0)	**20196**	1	1	0
B	(6, 1, 0, 0)	**16302**	1	1	0
B	(4, 1, 1, 1)	**12375**	1	1	0
B	(8, 0, 0, 0)	**9867**	1	1	0
B	(3, 2, 1, 0)	**9009**	1	1	0
B	(4, 2, 0, 0)	**8748**	1	1	0
B	(4, 1, 1, 0)	**7700**	2	2	0
B	(5, 1, 0, 0)	**7140**	3	3	0
B	(7, 0, 0, 0)	**5148**	1	1	0
B	(3, 1, 1, 1)	**3900**	3	3	0
B	(4, 1, 0, 0)	**2772**	6	6	0
B	(2, 2, 1, 1)	**2772**	1	1	0
B	(3, 2, 0, 0)	**2574**	3	3	0
B	(6, 0, 0, 0)	**2508**	2	2	0
B	(3, 1, 1, 0)	**2457**	7	7	0
B	(2, 2, 1, 0)	**1650**	2	2	0
B	(5, 0, 0, 0)	**1122**	3	3	0
B	(2, 1, 1, 1)	**924**	9	9	0
B	(3, 1, 0, 0)	**910**	12	12	0
B	(2, 1, 1, 0)	**594**	12	12	0
B	(2, 2, 0, 0)	**495**	6	6	0
B	(4, 0, 0, 0)	**450**	7	7	0
B	(2, 1, 0, 0)	**231**	19	19	0

Type	SO(9) Weight	Dim	Target Mult (R_n)	Recon Mult (from SO(10))	Difference
B	(3, 0, 0, 0)	**156**	11	11	0
B	(1, 1, 1, 1)	**126**	9	9	0
B	(1, 1, 1, 0)	**84**	13	13	0
B	(2, 0, 0, 0)	**44**	18	18	0
B	(1, 1, 0, 0)	**36**	17	17	0
B	(1, 0, 0, 0)	**9**	16	16	0
B	(0, 0, 0, 0)	**1**	8	8	0
F	(4.5, 1.5, 0.5, 0.5)	**27648**	1	1	0
F	(6.5, 0.5, 0.5, 0.5)	**27456**	1	1	0
F	(5.5, 0.5, 0.5, 0.5)	**12672**	2	2	0
F	(3.5, 1.5, 0.5, 0.5)	**9504**	4	4	0
F	(4.5, 0.5, 0.5, 0.5)	**5280**	5	5	0
F	(2.5, 1.5, 1.5, 0.5)	**5040**	1	1	0
F	(2.5, 2.5, 0.5, 0.5)	**4928**	1	1	0
F	(2.5, 1.5, 0.5, 0.5)	**2560**	10	10	0
F	(3.5, 0.5, 0.5, 0.5)	**1920**	11	11	0
F	(1.5, 1.5, 1.5, 0.5)	**768**	4	4	0
F	(2.5, 0.5, 0.5, 0.5)	**576**	20	20	0
F	(1.5, 1.5, 0.5, 0.5)	**432**	15	15	0
F	(1.5, 0.5, 0.5, 0.5)	**128**	27	27	0
F	(0.5, 0.5, 0.5, 0.5)	**16**	18	18	0

Table 10: Level 9 term-by-term SO(9) branching reconstruction verification (exact difference = 0).

A.10 Level $n = 10$ Representation Data

Single-Level Prefactor r_n (in SO(9)):

- **Bosons:** $1 \times$ **17875** (9, 0, 0, 0), $1 \times$ **33957** (7, 1, 0, 0), $1 \times$ **5148** (7, 0, 0, 0), $1 \times$ **46683** (6, 1, 1, 0), $2 \times$ **16302** (6, 1, 0, 0), $1 \times$ **2508** (6, 0, 0, 0), $1 \times$ **23868** (5, 2, 0, 0), $1 \times$ **32725** (5, 1, 1, 1), $1 \times$ **20196** (5, 1, 1, 0), $3 \times$ **7140** (5, 1, 0, 0), $4 \times$ **1122** (5, 0, 0, 0), $1 \times$ **31500** (4, 2, 1, 0), $2 \times$ **8748** (4, 2, 0, 0), $2 \times$ **12375** (4, 1, 1, 1), $5 \times$ **7700** (4, 1, 1, 0), $7 \times$ **2772** (4, 1, 0, 0), $5 \times$ **450** (4, 0, 0, 0), $1 \times$ **4004** (3, 3, 0, 0), $1 \times$ **15444** (3, 2, 1, 1), $2 \times$ **9009** (3, 2, 1, 0), $5 \times$ **2574** (3, 2, 0, 0), $7 \times$ **3900** (3, 1, 1, 1), $7 \times$ **2457** (3, 1, 1, 0), $12 \times$ **910** (3, 1, 0, 0), $11 \times$ **156** (3, 0, 0, 0), $3 \times$ **2772** (2, 2, 1, 1), $6 \times$ **1650** (2, 2, 1, 0), $5 \times$ **495** (2, 2, 0, 0), $9 \times$ **924** (2, 1, 1, 1), $15 \times$ **594** (2, 1, 1, 0), $17 \times$ **231** (2, 1, 0, 0), $10 \times$ **44** (2, 0, 0, 0), $10 \times$ **126** (1, 1, 1, 1), $10 \times$ **84** (1, 1, 1, 0), $13 \times$ **36** (1, 1, 0, 0), $12 \times$ **9** (1, 0, 0, 0), $3 \times$ **1** (0, 0, 0, 0)
- **Fermions:** $1 \times$ **54912** (7.5, 0.5, 0.5, 0.5), $1 \times$ **27456** (6.5, 0.5, 0.5, 0.5), $1 \times$ **68640** (5.5, 1.5, 0.5, 0.5), $3 \times$ **12672** (5.5, 0.5, 0.5, 0.5), $3 \times$ **27648** (4.5, 1.5, 0.5, 0.5), $6 \times$ **5280** (4.5, 0.5, 0.5, 0.5), $1 \times$ **24192** (3.5, 2.5, 0.5, 0.5), $1 \times$ **19712** (3.5, 1.5, 1.5, 0.5), $7 \times$ **9504** (3.5, 1.5, 0.5, 0.5), $11 \times$ **1920** (3.5, 0.5, 0.5, 0.5), $2 \times$ **4928** (2.5, 2.5, 0.5, 0.5), $4 \times$ **5040** (2.5, 1.5, 1.5, 0.5), $14 \times$ **2560** (2.5, 1.5, 0.5, 0.5), $19 \times$ **576** (2.5, 0.5, 0.5, 0.5), $2 \times$ **672** (1.5, 1.5, 1.5, 1.5), $5 \times$ **768** (1.5, 1.5, 1.5, 0.5), $15 \times$ **432** (1.5, 1.5, 0.5, 0.5), $22 \times$ **128** (1.5, 0.5, 0.5, 0.5), $12 \times$ **16** (0.5, 0.5, 0.5, 0.5)

Cumulative Spectrum $R_n = \sum_{l=1}^{n} r_l$ (in SO(9)):

- **Bosons:** $11 \times$ **1** (0, 0, 0, 0), $28 \times$ **9** (1, 0, 0, 0), $28 \times$ **44** (2, 0, 0, 0), $22 \times$ **156** (3, 0, 0, 0), $30 \times$ **36** (1, 1, 0, 0), $12 \times$ **450** (4, 0, 0, 0), $36 \times$ **231** (2, 1, 0, 0), $23 \times$ **84** (1, 1, 1, 0), $7 \times$ **1122** (5, 0, 0, 0), $24 \times$ **910** (3, 1, 0, 0), $27 \times$ **594** (2, 1, 1, 0), $19 \times$ **126** (1, 1, 1, 1), $3 \times$ **2508** (6, 0, 0, 0), $13 \times$ **2772** (4, 1, 0, 0), $14 \times$ **2457** (3, 1, 1, 0), $11 \times$ **495** (2, 2, 0, 0), $18 \times$ **924** (2, 1, 1, 1), $2 \times$ **5148** (7, 0, 0, 0), $6 \times$ **7140** (5, 1, 0, 0), $7 \times$ **7700** (4, 1, 1, 0), $8 \times$ **2574** (3, 2, 0, 0), $10 \times$ **3900** (3, 1, 1, 1), $8 \times$ **1650** (2, 2, 1, 0), $1 \times$ **9867** (8, 0, 0, 0), $3 \times$ **16302** (6, 1, 0, 0), $2 \times$ **20196** (5, 1, 1, 0), $3 \times$ **8748** (4, 2, 0, 0), $3 \times$ **12375** (4, 1, 1, 1), $3 \times$ **9009** (3, 2, 1, 0), $4 \times$ **2772** (2, 2, 1, 1), $1 \times$ **17875** (9, 0, 0, 0), $1 \times$ **33957** (7, 1, 0, 0), $1 \times$ **46683** (6, 1, 1, 0), $1 \times$ **23868** (5, 2, 0, 0), $1 \times$ **32725** (5, 1, 1, 1), $1 \times$ **31500** (4, 2, 1, 0), $1 \times$ **4004** (3, 3, 0, 0), $1 \times$ **15444** (3, 2, 1, 1)
- **Fermions:** $30 \times$ **16** (0.5, 0.5, 0.5, 0.5), $49 \times$ **128** (1.5, 0.5, 0.5, 0.5), $39 \times$ **576** (2.5, 0.5, 0.5, 0.5), $22 \times$ **1920** (3.5, 0.5, 0.5, 0.5), $30 \times$ **432** (1.5, 1.5, 0.5, 0.5), $11 \times$ **5280** (4.5, 0.5, 0.5, 0.5), $24 \times$ **2560** (2.5, 1.5, 0.5, 0.5), $5 \times$ **12672** (5.5, 0.5, 0.5, 0.5), $11 \times$ **9504** (3.5, 1.5, 0.5, 0.5), $9 \times$ **768** (1.5, 1.5, 1.5, 0.5), $2 \times$ **27456** (6.5, 0.5, 0.5, 0.5), $4 \times$ **27648** (4.5, 1.5, 0.5, 0.5), $3 \times$ **4928** (2.5, 2.5, 0.5, 0.5), $5 \times$ **5040** (2.5, 1.5, 1.5, 0.5), $1 \times$ **54912** (7.5, 0.5, 0.5, 0.5), $1 \times$ **68640** (5.5, 1.5, 0.5, 0.5), $1 \times$ **24192** (3.5, 2.5, 0.5, 0.5), $1 \times$ **19712** (3.5, 1.5, 1.5, 0.5), $2 \times$ **672** (1.5, 1.5, 1.5, 1.5)

SO(10) **Supermultiplet Decomposition:**

- **Bosons:** $5 \times$ **45** (1, 1, 0, 0, 0), $4 \times$ **210** (1, 1, 1, 1, 0), $1 \times$ **126** (1, 1, 1, 1, 1), $6 \times$ **320** (2, 1, 0, 0, 0), $3 \times$ **945** (2, 1, 1, 0, 0), $5 \times$ **1728** (2, 1, 1, 1, 0), $1 \times$ **770** (2, 2, 0, 0, 0), $3 \times$ **2970** (2, 2, 1, 0, 0), $2 \times$ **5940** (2, 2, 1, 1, 0), $1 \times$ **3696** (2, 2, 1, 1, 1), $5 \times$ **210** (3, 0, 0, 0, 0), $2 \times$ **1386** (3, 1, 0, 0, 0), $4 \times$ **4312** (3, 1, 1, 0, 0), $1 \times$ **8085** (3, 1, 1, 1, 0), $5 \times$ **4950** (3, 1, 1, 1, 1), $2 \times$ **4410** (3, 2, 0, 0, 0), $2 \times$ **17920** (3, 2, 1, 0, 0), $1 \times$ **23040** (3, 2, 1, 1, 1), $1 \times$ **7644** (3, 3, 0, 0, 0), $3 \times$ **660** (4, 0, 0, 0, 0), $1 \times$ **4608** (4, 1, 0, 0, 0), $4 \times$ **14784** (4, 1, 1, 0, 0), $2 \times$ **17325** (4, 1, 1, 1, 1), $1 \times$ **16380** (4, 2, 0, 0, 0), $1 \times$ **68640** (4, 2, 1, 0, 0), $2 \times$ **1782** (5, 0, 0, 0, 0), $1 \times$ **12870** (5, 1, 0, 0, 0), $1 \times$ **42120** (5, 1, 1, 0, 0), $1 \times$ **50050** (5, 1, 1, 1, 1), $1 \times$ **48510** (5, 2, 0, 0, 0), $1 \times$ **31680** (6, 1, 0, 0, 0), $1 \times$ **105105** (6, 1, 1, 0, 0), $1 \times$ **70785** (7, 1, 0, 0, 0), $1 \times$ **37180** (9, 0, 0, 0, 0)
- **Fermions:** $2 \times$ **16** (0.5, 0.5, 0.5, 0.5, 0.5), $8 \times$ **144** (1.5, 0.5, 0.5, 0.5, 0.5), $5 \times$ **560** (1.5, 1.5, 0.5, 0.5, 0.5), $4 \times$ **1200** (1.5, 1.5, 1.5, 0.5, 0.5), $2 \times$ **672** (1.5, 1.5, 1.5, 1.5, 1.5), $8 \times$ **720** (2.5, 0.5, 0.5, 0.5, 0.5), $7 \times$ **3696** (2.5, 1.5, 0.5, 0.5, 0.5), $4 \times$ **8800** (2.5, 1.5, 1.5, 0.5, 0.5), $2 \times$ **8064** (2.5, 2.5, 0.5, 0.5, 0.5), $5 \times$ **2640** (3.5, 0.5, 0.5, 0.5, 0.5), $5 \times$ **15120** (3.5, 1.5, 0.5, 0.5, 0.5), $1 \times$ **38016** (3.5, 1.5, 1.5, 0.5, 0.5), $1 \times$ **43680** (3.5, 2.5, 0.5, 0.5, 0.5), $3 \times$ **7920** (4.5, 0.5, 0.5, 0.5, 0.5), $3 \times$ **48048** (4.5, 1.5, 0.5, 0.5, 0.5), $2 \times$ **20592** (5.5, 0.5, 0.5, 0.5, 0.5), $1 \times$ **129360** (5.5, 1.5, 0.5, 0.5, 0.5), $1 \times$ **48048** (6.5, 0.5, 0.5, 0.5, 0.5), $1 \times$ **102960** (7.5, 0.5, 0.5, 0.5, 0.5)

SO(9) **Term-by-Term Branching Reconstruction Verification:**

Type	SO(9) Weight	Dim	Target Mult (R_n)	Recon Mult (from SO(10))	Difference
B	(6, 1, 1, 0)	**46683**	1	1	0
B	(7, 1, 0, 0)	**33957**	1	1	0

Type	SO(9) Weight	Dim	Target Mult (R_n)	Recon Mult (from SO(10))	Difference
B	(5, 1, 1, 1)	**32725**	1	1	0
B	(4, 2, 1, 0)	**31500**	1	1	0
B	(5, 2, 0, 0)	**23868**	1	1	0
B	(5, 1, 1, 0)	**20196**	2	2	0
B	(9, 0, 0, 0)	**17875**	1	1	0
B	(6, 1, 0, 0)	**16302**	3	3	0
B	(3, 2, 1, 1)	**15444**	1	1	0
B	(4, 1, 1, 1)	**12375**	3	3	0
B	(8, 0, 0, 0)	**9867**	1	1	0
B	(3, 2, 1, 0)	**9009**	3	3	0
B	(4, 2, 0, 0)	**8748**	3	3	0
B	(4, 1, 1, 0)	**7700**	7	7	0
B	(5, 1, 0, 0)	**7140**	6	6	0
B	(7, 0, 0, 0)	**5148**	2	2	0
B	(3, 3, 0, 0)	**4004**	1	1	0
B	(3, 1, 1, 1)	**3900**	10	10	0
B	(4, 1, 0, 0)	**2772**	13	13	0
B	(2, 2, 1, 1)	**2772**	4	4	0
B	(3, 2, 0, 0)	**2574**	8	8	0
B	(6, 0, 0, 0)	**2508**	3	3	0
B	(3, 1, 1, 0)	**2457**	14	14	0
B	(2, 2, 1, 0)	**1650**	8	8	0
B	(5, 0, 0, 0)	**1122**	7	7	0
B	(2, 1, 1, 1)	**924**	18	18	0
B	(3, 1, 0, 0)	**910**	24	24	0
B	(2, 1, 1, 0)	**594**	27	27	0
B	(2, 2, 0, 0)	**495**	11	11	0
B	(4, 0, 0, 0)	**450**	12	12	0
B	(2, 1, 0, 0)	**231**	36	36	0
B	(3, 0, 0, 0)	**156**	22	22	0
B	(1, 1, 1, 1)	**126**	19	19	0
B	(1, 1, 1, 0)	**84**	23	23	0
B	(2, 0, 0, 0)	**44**	28	28	0
B	(1, 1, 0, 0)	**36**	30	30	0
B	(1, 0, 0, 0)	**9**	28	28	0
B	(0, 0, 0, 0)	**1**	11	11	0
F	(5.5, 1.5, 0.5, 0.5)	**68640**	1	1	0
F	(7.5, 0.5, 0.5, 0.5)	**54912**	1	1	0
F	(4.5, 1.5, 0.5, 0.5)	**27648**	4	4	0
F	(6.5, 0.5, 0.5, 0.5)	**27456**	2	2	0
F	(3.5, 2.5, 0.5, 0.5)	**24192**	1	1	0
F	(3.5, 1.5, 1.5, 0.5)	**19712**	1	1	0
F	(5.5, 0.5, 0.5, 0.5)	**12672**	5	5	0
F	(3.5, 1.5, 0.5, 0.5)	**9504**	11	11	0
F	(4.5, 0.5, 0.5, 0.5)	**5280**	11	11	0
F	(2.5, 1.5, 1.5, 0.5)	**5040**	5	5	0
F	(2.5, 2.5, 0.5, 0.5)	**4928**	3	3	0
F	(2.5, 1.5, 0.5, 0.5)	**2560**	24	24	0
F	(3.5, 0.5, 0.5, 0.5)	**1920**	22	22	0
F	(1.5, 1.5, 1.5, 0.5)	**768**	9	9	0

Type	SO(9) Weight	Dim	Target Mult (R_n)	Recon Mult (from SO(10))	Difference
F	(1.5, 1.5, 1.5, 1.5)	**672**	2	2	0
F	(2.5, 0.5, 0.5, 0.5)	**576**	39	39	0
F	(1.5, 1.5, 0.5, 0.5)	**432**	30	30	0
F	(1.5, 0.5, 0.5, 0.5)	**128**	49	49	0
F	(0.5, 0.5, 0.5, 0.5)	**16**	30	30	0

Table 11: Level 10 term-by-term SO(9) branching reconstruction verification (exact difference = 0).

References

[1] I. Bars, *Stringy Evidence for D=11 Structure in Strongly Coupled Type II-A Superstring*, Phys. Rev. Lett. **74** (1995) 3244, arXiv:hep-th/9503228.